

$$1) P[y > 0] = P[x_1 x_2 > 0] = P[x_1 > 0, x_2 > 0] + P[x_1 < 0, x_2 < 0]$$

$$\stackrel{\uparrow}{=} P[x_1 > 0] P[x_2 > 0] + P[x_1 < 0] \cdot P[x_2 < 0]$$

indep.

$$f_x(e) \text{ e symmetric} \Rightarrow P[x_1 > 0] = \frac{1}{2}$$

$$P[y > 0] = \frac{1}{2}$$

$$2) P_e = 0.1 \cdot P\left[n_k > \frac{V_0}{2}\right] + 0.1 P\left[n_k < -\frac{V_0}{2}\right]$$

$$+ 0.8 P\left[|n_k| > \frac{V_0}{2}\right]$$

$$= 0.2 Q\left(\frac{V_0}{\sigma_n \cdot 2}\right) + 0.8 \cdot 2 \cdot Q\left(\frac{V_0}{\sigma_n \cdot 2}\right)$$

$$= 1.8 Q(50)$$

$$3) \bar{R}_w(f) = \left| \frac{1}{T} G(f) \right|^2 \cdot R_e(f); \quad \begin{matrix} m_e = 0 \\ \sigma_e^2 = 0.4 \end{matrix}$$

$$R_e(f) = \sigma_e^2 \cdot T = 0.4T$$

$$G(f) = V_0 \cdot \int_0^{+\infty} e^{-(\frac{5}{T} + j2\pi f)t} dt$$

$$= \frac{V_0}{\frac{5}{T} + j2\pi f}$$

$$|G(f)|^2 = \frac{V_0^2}{\frac{25}{T^2} + 4\pi^2 f^2}$$

$$\bar{R}_w(f) = \frac{V_0^2}{25 + 4\pi^2 T^2 f^2} \cdot 0.4T$$

$$\bar{P}_w = \int_{-\infty}^{+\infty} \bar{R}_w(f) df = 0.04 V_0^2$$

4.

$$L(f) = \frac{\frac{1}{j\bar{u}fC}}{R + \frac{1}{j\bar{u}fC}} = \frac{1}{1 + j\bar{u}fRC}$$

Porto

$$H(f) = h_0 \cdot (1 + j\bar{u}fRC) \cdot \text{rect} \frac{f \cdot T}{(1+\alpha)}$$

in be

$$C(f) = G(f) \cdot L(f) \cdot H(f) = h_0 \cdot G(f)$$

con esdamento e coseno nel beito $\Rightarrow$ no isi

$$\sigma^2 = \int_{-\infty}^{+\infty} R_0 |H(f)|^2 df = \int_{-(1+\alpha)f_N}^{(1+\alpha)f_N} R_0 \cdot h_0^2 (1 + 4\pi^2 f^2 (RC)^2) df$$

$$= R_0 h_0^2 \frac{(1+\alpha)}{T} + R_0 h_0^2 4\pi^2 (RC)^2 \cdot \frac{1}{3} \cdot 2 (1+\alpha)^3 f_N^3$$

5.

$$m_c(hT) = T \sum_{k=0}^{+\infty} 0.5^k n(kT - kT) \quad m_{m_c} = 0$$

$$R_{mc}(f) = R_m(f) \cdot |H(f)|^2 = R_0 |H(f)|^2$$

$$\begin{aligned} R_{mc} &= \int_0^{\frac{1}{T}} R_{mc}(f) df = \int_0^{\frac{1}{T}} R_0 |H(f)|^2 df = R_0 \cdot T \cdot \sum_{k=0}^{+\infty} h^2(kT) \\ &= R_0 T \sum_{k=0}^{+\infty} ((0.5)^2)^k = \frac{R_0 T}{1 - \frac{1}{4}} = \frac{4}{3} R_0 T \end{aligned}$$